Selecting Counterfactuals for Patient-Safety Events in Multimodal Electronic Health Records

A structural diagnostic-process model and a sequential risk-set matching procedure

Working draft

June 5, 2026

Abstract

We study how to construct counterfactual comparison units for a small set of patient-safety–indicator (PSI) events ($n_{\mathcal{C}} = 145$) against a very large donor pool ($\sim 51 \times 10^6$ admissions) of multimodal electronic-health-record (EHR) data. We first write down a *structural model of the diagnostic process* that treats the observed record as a selected, action-dependent projection of a latent clinical state through the clinician's information-acquisition and treatment policy. The model makes explicit why "missingness" in EHR is informative, why exchangeability must condition on the evolving *history* rather than the current snapshot, and why the appropriate low-dimensional summary for matching is a *balancing score* rather than a variance-maximizing projection. We further formalize a *dynamic positivity* problem: on the stay clock the propensity score behaves as a Doob martingale whose mean is pinned at the event rate while its variance grows, so the case and control score distributions separate and common support is extinguished by the time the event is registered — which recasts the blanking window as the knob that trades unconfoundedness against positivity. We then give a staged selection procedure: (i) baseline coarsened exact matching (CEM) on time-invariant and admission covariates with a missingness-as-information rule; (ii) a time-varying, landmarked *risk-set* matching step in which the multimodal history at each time is reduced through a large-scale propensity / prognostic score and matched within the event-free risk set; and (iii) a verification stage that uses a placebo-outcome causal forest to falsify residual selection. We close with diagnostics (balance, negative controls, empirical calibration, sensitivity) and a list of open problems.

1 Introduction

We observe a target set $\mathcal{C}$ of $n_{\mathcal{C}} = |\mathcal{C}| = 145$ inpatient admissions flagged for a specific patient-safety event (a "PSI-positive" admission), and a donor pool $\mathcal{D}$ of eligible event-free admissions drawn from $\sim 51 \times 10^6$ patients (the OMNY data are described in Section 2). We restrict attention to methods that *select, weight, or synthesize comparison observations* under an unconfoundedness-type assumption, in the sense of the accompanying review.

The defining features of the problem are: (a) the target group is *rare* (145 events) while the donor pool is enormous, so variance, not data scarcity, is the binding constraint; (b) the data are *multimodal* (numeric labs and vitals; high-dimensional sparse code sets for procedures, prescriptions, and diagnoses; free text); (c) covariates are *time-varying* on a discrete grid anchored at admission; and (d) each case carries an *event time*, which the donors do not, creating an alignment problem that, handled naively, produces immortal-time bias.

What "counterfactual" means here. A PSI is an *outcome*, not a manipulable treatment, so there is no counterfactual "to the event" in the potential-outcomes sense. The object we construct is, for each case, a matched set of event-free admissions that were *exchangeable with the case up to a landmark time*; their subsequent trajectory stands in for the case's counterfactual "business-as-usual" continuation. Section 4 states this estimand precisely. If instead the target is the effect of a specific upstream *care action* on PSI risk, that action becomes the treatment and the active-comparator, new-user logic of the review applies with only notational changes; we flag those points as they arise.

Contributions. The paper makes four connected contributions. First, a *structural model of the diagnostic process* (Section 3) that derives, rather than assumes, why EHR missingness is informative and why comparison observations must be selected on the evolving history. Second, an *econometric identification* of the counterfactual continuation that makes the vector of individual unobservables explicit and states selection-on-observables as a proxy condition on the history (Section 4). Third, a *dynamic-positivity* result (Section 5): the propensity score is a Doob martingale whose variance grows along the stay, so common support is a moving target that is extinguished by the time the event is registered, which gives the blanking window its meaning. Fourth, an operational *staged selection procedure* (Section 6) with data-governance filters, missingness-as-information matching, a balancing/prognostic-score reduction of the multimodal history, and a placebo-falsification verification stage, together with the diagnostics needed to defend it (Section 7).

Roadmap. Section 2 fixes the data and notation; Section 3 gives the structural model; Section 4 states the estimand and identification; Section 5 develops the information–positivity trade-off; Section 6 gives the procedure; Sections 7–9 cover diagnostics, positioning relative to existing designs, and open problems.

2 Setup, data, and notation

Data. The data follow the OMNY common structure: a central `ENCOUNTERS` table keyed on (patient, encounter), with encounter-level child tables for diagnoses, labs, vitals, procedures, prescription orders and administrations, payers, scores, and free-text notes; longitudinal problem lists keyed on patient only; and a broader claims history that does not join to the encounter tables. Demographics live on `ENCOUNTERS` rather than in a separate patient table, and several streams are sparse or supplier-dependent — which forces the governance and quality filters of Section 6.

We index admissions by i. The target ("case") set is $\mathcal{C}$ and the donor set is $\mathcal{D}$. Two clocks must be kept distinct.

- **Stay clock.** A discrete grid $t \in \{0, 1, 2, \dots\}$ with spacing $\Delta = 4$h, so wall time since admission is $t\,\Delta$. Admission is $t = 0$; $t = 1$ is 4h after admission, $t = 2$ is 8h after admission, and so on.[1]
- **Event clock.** For a case $i \in \mathcal{C}$, let E_i (in grid units) be the time the PSI is registered, and let b be a *blanking window* (here $b = 6$, i.e. 24h). The *landmark* (index) time is

$$t_i^* \;=\; E_i - b,$$

  and the trajectory we use for case i is its history on $\{0, 1, \dots, t_i^*\}$. The blanking window keeps the event's physiological prodrome out of the predictors and is a sensitivity parameter (Section 9).

[1]This corrects an ambiguity in the original specification, which read "t=2 is 8h *before* admission"; the grid runs forward from admission.

Covariates. $X_i \in \mathbb{R}^{d_X}$ collects *time-invariant* attributes and admission-time records: demographics (age band, sex, race/ethnicity), employment, facility type / size / urban–rural, admission department, baseline anthropometry (height, weight, BMI), baseline comorbidity burden from the problem list, and (descriptively) SDOH. Time-varying multimodal covariates at time t are stacked by modality,

$$L_{it} \;=\; \big(\,L_{it}^{\text{lab}},\; L_{it}^{\text{vit}},\; L_{it}^{\text{rx}},\; L_{it}^{\text{px}}\,\big),$$

for labs, vitals, prescriptions (orders and administrations), and procedures, respectively. Histories are written with an overbar, $\overline{L}_{it} = (L_{i0}, \ldots, L_{it})$, and likewise $\overline{A}_{it}$ for actions.

The diagnosis stream is held apart. Recorded diagnoses and problem-list updates are *not* part of L_{it}; we give them their own symbol,

$$\gamma_{it} \;=\; \big(\text{ICD-coded diagnoses and problem-list updates active at } t\big), \qquad \overline{\gamma}_{it} = (\gamma_{i0}, \ldots, \gamma_{it}),$$

because they are categorically different from the physiologic covariates in L_{it}. Whereas L_{it} records measurements of the patient, γ_{it} records the clinician's *labels* — the output f_D of the diagnostic process (Section 3). Diagnoses therefore sit closest to the outcome on the causal path: indeed the PSI itself is an entry in this very stream, so γ and the outcome Y are the same kind of object, with Y its terminal, qualifying realization. Tracking γ_{it} separately lets us reason explicitly about *when* conditioning on it is legitimate — baseline comorbidity and stable problems are bona fide confounders — and when it becomes outcome-descendant, namely the pre-event diagnostic cascade that the blanking window of Section 5 is designed to exclude.

Observation versus value. For each measurable item k (e.g. a specific lab analyte) we separate the *act of observing* from the *observed value*:

$$R_{itk}^{\text{lab}} \in \{0,1\} \quad (1 \text{ iff item } k \text{ ordered/observed at } t), \qquad M_{itk}^{\text{lab}} \text{ observed iff } R_{itk}^{\text{lab}} = 1.$$

Thus $L_{it}^{\text{lab}} = R_{it}^{\text{lab}} \odot M_{it}^{\text{lab}}$, and the order pattern R_{it} is itself a covariate generated by the diagnostic process (Section 3). This is the notational device that lets "not measured" be treated as information rather than as a hole. The diagnosis stream γ_{it} admits no analogous value/observation split: it is a pure label that is either recorded or not, and its *absence* is informative in the same way an un-ordered lab is.

Latent state, actions, outcome. $H_{it} \in \mathcal{H}$ is the latent clinical state; $A_{it} = (O_{it}, A_{it}^{\text{rx}}, A_{it}^{\text{px}})$ collects all care actions — test/procedure *orders* O_{it} and *treatments* $A_{it}^{\text{rx}}, A_{it}^{\text{px}}$; $Y_{it} \in \{0,1\}$ indicates a PSI at t, with hazard $\lambda_{it} = \mathbb{P}(Y_{it} = 1 \,|\, \cdot)$. For a case, $Y_{i,E_i} = 1$. The counterfactual "control" regime, written (0), is the continuation *absent the event-precipitating pathway*; it is a target-defined regime rather than a manipulable assignment (Section 1), and the associated potential trajectories are $\{Y_{it}(0), \overline{L}_{it}(0)\}$.

Observed history (the information set). We collect everything observed through time t into the *information set*

$$\overline{\mathcal{I}}_{it} \;=\; \big(X_i,\; \overline{L}_{it},\; \overline{\gamma}_{it},\; \overline{A}_{it}\big),$$

the object the clinician conditions on in Section 3, the object we condition on for identification (Section 4), and the object we reduce to a scalar balancing score (Sections 5–6). For a case, histories are truncated at the landmark: we use $\overline{\mathcal{I}}_{it}$ only for $t \le t_i^*$.

Nested (append-only) information. Each entry of $\overline{\mathcal{I}}_{it}$ is a history — a prefix that the next period extends — so information accumulates by construction. Writing Δ_{it} for the bundle of values, orders, results, and codes newly *available* in window t,

$$\overline{\mathcal{I}}_{it} \;=\; \big(\overline{\mathcal{I}}_{i,t-1},\; \Delta_{it}\big), \qquad \overline{\mathcal{I}}_{i0} = \big(X_i,\; \text{admission record}\big),$$

so $\overline{\mathcal{I}}_{it}$ contains $\overline{\mathcal{I}}_{i,t-1}$ as a sub-record.

Definition 1 (Nested information)**.** The observed data form an *append-only* record: the period-t increment Δ_{it} is only ever added to the information set, never overwritten or removed, and each datum is time-stamped by *availability* (specimen/result time, not order time) so that it enters $\overline{\mathcal{I}}_{it}$ only once it is known. The policies are adapted — O_{it} and $(A^{\mathrm{rx}}_{it}, A^{\mathrm{px}}_{it})$ depend only on $\overline{\mathcal{I}}_{it}$, with no anticipation of values not yet observed. Consequently $\mathcal{F}_{it} = \sigma(\overline{\mathcal{I}}_{it})$ is an increasing filtration,

$$\mathcal{F}_{i,t-1} \;\subseteq\; \mathcal{F}_{it} \qquad \text{for every } t,$$

with equality in windows where $\Delta_{it} = \varnothing$: at $t+1$ there is *at least* as much information as at t.

This construction makes $(\mathcal{F}_{it})_t$ a filtration in the probabilistic sense and underwrites the Doob-martingale argument of Section 5. It is distinct from the Markov property of the latent state: H_{it} is memoryless by (1), but the *observer's* information is not — the state may rise or fall while knowledge of the path only grows.

Remark 1 (Keep nesting at the information level, not the feature level)**.** Nesting holds for the raw history, not for an arbitrary fixed-width summary of it. A snapshot feature such as "most recent value" overwrites its predecessor, so $\sigma(\text{feature}_{t+1}) \not\supseteq \sigma(\text{feature}_t)$ in general, and the balancing score e_t is a t-dependent coarsening of $\mathcal{F}_{it}$ whose own σ-algebra need not nest either. We therefore conduct identification and the martingale argument on the cumulative filtration $\mathcal{F}_{it}$ and treat e_t, ψ_t as measurable functions of it. If the engineered feature vector must itself be nested — as in an online or sequential estimator — use *cumulative* summaries (running counts, running min/max, ever-abnormal indicators, cumulative sums, time-since-last-update) that retain rather than discard the past, and never let a late-arriving or amended record — a backfilled lab, a retroactive diagnosis code, a post-discharge claim — re-date itself into an earlier $\overline{\mathcal{I}}_{it}$.

Selection objects. $\kappa(\cdot)$ is a coarsening operator and $\mathcal{S}$ the induced strata (Section 6.2); $e_t(\cdot)$ and $\psi_t(\cdot)$ are the time-varying propensity and prognostic scores (Section 6.3); $\mathcal{M}_i(t) \subset \mathcal{D}$ is the matched donor set for case i at landmark t.

3 A structural model of the diagnostic process

We model the observed multimodal record as the joint output of (i) a latent patient trajectory and (ii) the clinician's sequential information-acquisition and treatment policy. The model is a discrete-time structural causal system on the stay clock.

Patient type and latent dynamics. A latent admission "type" U_i generates time-invariant attributes and the initial state,

$$X_i = f_X(U_i, \varepsilon_i^X), \qquad H_{i0} = f_{H0}(X_i, \varepsilon_{i0}^H),$$

and the latent state evolves as a controlled Markov process,

$$H_{i,t+1} \;=\; f_H\big(H_{it},\, A_{it},\, X_i,\, \varepsilon^H_{i,t+1}\big). \tag{1}$$

Diagnostic policy (information acquisition). Recall the information set $\overline{\mathcal{I}}_{it}$ of Section 2 — the observed history available when the period-t decisions are made. The diagnostic process is the ordering policy

$$O_{it} \;=\; f_O\big(\overline{\mathcal{I}}_{it};\, \theta_O\big), \qquad R_{itk} = \mathbb{1}\{\text{item } k \in O_{it}\}, \tag{2}$$

which determines *which* labs and procedures are ordered and hence *which* items are observed. Equation (2) is the engine of informative observation: because O_{it} depends on $\overline{\mathcal{I}}_{it}$, which is driven by the latent state through past measurements, the data are missing-not-at-random by construction.

Measurement and treatment. Observed values are noisy reflections of the latent state, realized only when ordered,

$$M_{itk} = f_M\big(H_{it}, k, \varepsilon^M_{itk}\big) \;\; \text{for } R_{itk} = 1, \tag{3}$$

and treatments follow a policy on the (now-updated) information set,

$$\big(A^{\text{rx}}_{it},\, A^{\text{px}}_{it}\big) = f_A\big(\overline{\mathcal{I}}_{it};\, \theta_A\big). \tag{4}$$

Diagnoses are noisy, coding-mediated labels of the latent state and information set, $\gamma_{it} = f_D(H_{it}, \overline{\mathcal{I}}_{it}, \varepsilon^D_{it})$.

Outcome. The PSI hazard depends on the latent state and the realized care history,

$$\lambda_{it} \;=\; \mathbb{P}\big(Y_{it} = 1 \,|\, H_{it},\, \overline{A}_{i,t-1},\, X_i\big) \;=\; h\big(H_{it}, \overline{A}_{i,t-1}, X_i\big). \tag{5}$$

Remark 2 (Three structural lessons)**.** Equations (1)–(5) carry the design implications. (1) *Missingness is a covariate.* "Item k not measured at t" is the event $R_{itk} = 0$, produced by f_O; it is informative about H_{it} and must be carried, not imputed away. (2) *Condition on history, not snapshots.* Both the care policy and the hazard depend on $\overline{\mathcal{I}}_{it}$; exchangeability therefore requires conditioning on the evolving observed history $\overline{\mathcal{I}}_{it}$, not the current snapshot — the time-varying-confounding setting. (3) *Reduce via a balancing score.* The right one-dimensional summary of the high-dimensional history is a function that renders case/donor assignment ignorable (a propensity-type *balancing score*) or that predicts the control hazard (a *prognostic score*) — not a variance-maximizing projection such as unsupervised PCA, which can discard precisely the outcome-relevant signal.

4 Estimand and identification

Fix a case $i \in \mathcal{C}$ with landmark t^*_i. We seek its counterfactual control continuation,

$$\tau_i \;=\; \big\{\, Y_{i,s}(0),\ \overline{L}_{i,s}(0) \;:\; s > t^*_i \,\big\},$$

estimated by the post-landmark trajectory of a matched donor set $\mathcal{M}_i(t^*_i)$. Aggregating over $\mathcal{C}$ yields a target-population (case) quantity — for example a divergence $\mathbb{E}_{\mathcal{C}}\big[\, Y_{i,s} - \widehat{Y}_{i,s}(0) \,\big]$ between factual cases and their estimated counterfactuals at horizon s.

Econometric setup: a vector of individual unobservables. Following the program-evaluation tradition, we make the individual heterogeneity explicit. Each admission carries a vector of *time-invariant unobserved values* $\alpha_i \in \mathbb{R}^{d_\alpha}$ — frailty, physiologic reserve, unmeasured baseline severity, genetic risk, care-seeking propensity — that is not contained in the observed baseline X_i. The latent admission type U_i of Section 3 thus splits into an observed projection $X_i = f_X(U_i, \cdot)$ and an unobserved remainder α_i, which propagates into the latent state through the dynamics (1). Writing the control potential outcome as a latent-index refinement of the hazard (5),

$$Y_{it}(0) = \mathbb{1}\left\{ m_t(\alpha_i, \overline{\mathcal{I}}_{i,t-1}) + \varepsilon_{it} > 0 \right\}, \qquad \mathbb{E}\left[\varepsilon_{it} \mid \alpha_i, \overline{\mathcal{I}}_{i,t-1}\right] = 0,$$

case status C_i is itself a functional of the same latent trajectory, hence of α_i. The identification problem is the classical one: α_i enters both the outcome and the selection rule, so any contrast that fails to account for it confounds case status with α_i ("selection on unobservables").

Assumption 1 (Landmark alignment / no immortal time)**.** A donor j is eligible to match case i at t_i^* only if j is still admitted and event-free at stay-time t_i^*, i.e. j is in the *risk set* $\mathcal{R}(t_i^*)$.

Assumption 2 (Sequential exchangeability given observed history)**.** At each landmark, conditional on the observed history, being on the to-become-a-case path is independent of the counterfactual control trajectory:

$$\left\{Y_{i,s}(0)\right\}_{s>t} \perp\!\!\!\perp \mathbb{1}\{i \in \mathcal{C}\} \;\Big|\; \overline{\mathcal{I}}_{it}.$$

Assumption 3 (Selection on observables despite α_i)**.** The observed multimodal history is informationally sufficient for the unobserved heterogeneity, in that selection no longer depends on α_i once the history is conditioned on:

$$C_i \perp\!\!\!\perp \alpha_i \;\Big|\; \overline{\mathcal{I}}_{it}.$$

Together with the latent-index model above, this delivers the exchangeability of Assumption 2: the longitudinal record in $\overline{\mathcal{I}}_{it}$ — repeated noisy measurements of the latent state driven by α_i — proxies α_i well enough that no residual α_i-channel drives differential selection. This is the econometric microfoundation of selection on observables, and it is a strong assumption: it asserts that the rich history substitutes for a within transformation.

Assumption 4 (Positivity / overlap)**.** For every history value with positive case density there is positive donor density in $\mathcal{R}(t)$: $0 < e_t(\overline{\mathcal{I}}_{it}) < 1$ on the support of the cases.

Assumption 5 (Consistency / SUTVA)**.** Trajectories are well defined for the regime each unit follows, with no interference across admissions. (Interference is plausible within a unit/facility — staffing, outbreaks — and is treated as a limitation in Section 9.)

Remark 3 (Latent-state caveat)**.** Assumption 2 conditions on the *observed* history $\overline{\mathcal{I}}_{it}$, not on the latent H_{it} of Section 3. Residual confounding by the unobserved component of H_{it} is therefore expected and is addressed by negative-control and sensitivity analysis (Section 7), in the spirit of the OHDSI program. Because all histories are evaluated at the landmark $t \leq t_i^*$, the diagnosis history $\overline{\gamma}_{it}$ entering the conditioning set excludes the pre-event cascade by construction; conditioning on $\overline{\gamma}$ beyond the landmark would condition on a descendant of the outcome (Section 5).

Remark 4 (Why α_i is proxied, not differenced out)**.** In a panel with an additively separable individual effect, α_i could be eliminated by a within (fixed-effects) or difference-in-differences transformation rather than proxied. That route is unavailable here: each case contributes a single right-censored spell, with no post-event "control" period and no pre-period in which the case is untreated, so there is nothing to difference against. Identification therefore rests on the proxy condition (Assumption 3), and its credibility is gauged by the sensitivity analyses of Sections 7 and 9 — negative controls, empirical calibration, the E-value, and Rosenbaum bounds — which quantify how strong a residual α_i-channel would have to be to overturn a finding.

Proposition 1 (Identification of the counterfactual continuation)**.** Under Assumptions 1–5, for each case i and horizon $s > t_i^*$ the counterfactual control outcome is identified by the risk-set comparison at the landmark,

$$\mathbb{E}\big[Y_{i,s}(0) \mid \overline{\mathcal{I}}_{i,t_i^*}\big] \;=\; \mathbb{E}\big[\,Y_{j,s} \;\big|\; j \in \mathcal{R}(t_i^*),\; \overline{\mathcal{I}}_{j,t_i^*} = \overline{\mathcal{I}}_{i,t_i^*}\big],$$

so that τ_i is recovered by matching or balancing donors to case i on $\overline{\mathcal{I}}_{i,t_i^*}$ — equivalently, by the balancing-score theorem, on the scalar $e_{t_i^*}(\overline{\mathcal{I}}_{i,t_i^*})$ — within the risk set. Identification requires the blanking window to be large enough that Assumption 4 holds at t_i^*; Section 5 shows this is the binding constraint.

5 Overlap is a moving target: the information–positivity trade-off

The identification conditions of Section 4 are often read as if conditioning on more of the record were unambiguously beneficial. In a longitudinal diagnostic setting they pull against each other, and the tension is *dynamic*: the further along the stay clock we condition, the more credible sequential exchangeability (Assumption 2) becomes and the less credible positivity (Assumption 4) becomes. The counterfactual group is therefore not a fixed set but a region that contracts as information accumulates.

Three regimes. At the no-information limit $t \to -\infty$ the balancing requirement is vacuous: we condition on no variable, every eligible admission qualifies as a comparison, and overlap is complete — but the comparison is maximally confounded. At admission, $t = 0$, conditioning on the baseline covariates X_i restricts the comparison to baseline-similar admissions; the admissible set is already much smaller, though overlap is still ample. As $t \to E_i$, conditioning on the full history — in particular on the test or exam that has by then revealed the impending PSI — leaves essentially the case itself: the comparison set is empty and positivity fails.

A martingale view of the propensity path. Let $C_i = \mathbb{1}\{i \text{ becomes a case during the stay}\}$ with marginal rate $\pi = \mathbb{P}(C_i = 1)$ (here $\pi \approx 145/(51 \times 10^6)$), let $\mathcal{F}_{it} = \sigma(\overline{\mathcal{I}}_{it})$ be the increasing filtration generated by the information set (increasing by construction, Definition 1), and let

$$e_{it} = \mathbb{P}\big(C_i = 1 \,|\, \mathcal{F}_{it}\big) = e_t(\overline{\mathcal{I}}_{it})$$

be the propensity path along the stay clock — the same score that the procedure (Section 6) estimates, there additionally restricted to the risk set $\mathcal{R}(t)$.

Proposition 2 (Dynamic positivity)**.** Under the increasing filtration $(\mathcal{F}_{it})_t$:

(i) $(e_{it})_t$ is a martingale with constant mean $\mathbb{E}[e_{it}] = \pi$ for every t;

(ii) $\mathrm{Var}(e_{it})$ is non-decreasing in t, rising from 0 under the trivial filtration toward the maximal $\pi(1-\pi)$ as $\mathcal{F}_{it}$ comes to determine C_i;

(iii) consequently the common-support mass $\omega(t) = \mathbb{P}\big(0 < e_{it} < 1 \,|\, C_i = 1\big)$ tends to contract, with $\omega(t) \to 1$ as $t \to -\infty$ and $\omega(t) \to 0$ as $t \to E_i$.

Sketch. (i) is the Lévy–Doob martingale $\mathbb{E}[C_i \,|\, \mathcal{F}_{it}]$ on an increasing filtration. (ii) follows from the law of total variance, $\mathrm{Var}(e_{it}) = \pi(1-\pi) - \mathbb{E}[\mathrm{Var}(C_i \,|\, \mathcal{F}_{it})]$, since the residual conditional variance is non-increasing as the filtration refines. (iii) is heuristic: a $[0,1]$-valued score whose variance approaches $\pi(1-\pi)$ is close to Bernoulli(π), so its mass concentrates at the endpoints — cases pile near 1, controls near 0 — and the region where both groups retain positive density shrinks. □

The mean of the score is pinned at the (tiny) event rate throughout; what changes is its *dispersion*. Early in the stay almost all admissions share nearly the same small score and are mutually comparable; late in the stay the score is bimodal and the two groups no longer occupy a common region. This is the formal content of the original intuition: at $t \to -\infty$ everyone is a potential counterfactual, by $t = 0$ the set has shrunk, and by the eve of the event it is empty.

The combinatorial counterpart: nested donor sets under MIB. The same contraction appears in finite samples, with no probability model, when matching is done by coarsened exact matching — a member of the monotonic-imbalance-bounding (MIB) class [6]. Fix the coarsening of each covariate (fixed bin edges) and, for a case i matched on the information available at t, let

$$\mathcal{M}_i(t) = \{\, l \in \mathcal{R}(t) \;:\; C(\overline{\mathcal{I}}_{lt}) = C(\overline{\mathcal{I}}_{it}) \,\}$$

be the donors in i's coarsened cell that are still at risk, where $C(\cdot)$ is the coarsening.

Proposition 3 (Monotone donor shrinkage)**.** Under an append-only information set (Definition 1) and fixed coarsening,

$$\mathcal{M}_i(t+1) \;\subseteq\; \mathcal{M}_i(t), \qquad \text{hence} \qquad |\mathcal{M}_i(t+1)| \;\leq\; |\mathcal{M}_i(t)|.$$

Proof. Coarsening is coordinate-wise with fixed bins, so $C(\overline{\mathcal{I}}_{i,t+1}) = \big(C(\overline{\mathcal{I}}_{it}),\, C(\Delta_{i,t+1})\big)$. A donor matches i at $t+1$ iff it matches on the period-t cell *and* on the period-$(t+1)$ increment, whence $\mathcal{M}_i(t+1) = \mathcal{M}_i(t) \cap \{\, l : C(\Delta_{l,t+1}) = C(\Delta_{i,t+1}) \,\} \subseteq \mathcal{M}_i(t)$. Risk-set attrition $\mathcal{R}(t+1) \subseteq \mathcal{R}(t)$ — no re-entry within a spell — only reinforces the inclusion. □

The donor set is thus monotonically non-increasing in t: constant in windows that add no discriminating covariate ($\Delta_{it} = \varnothing$, or a covariate constant within the cell) and strictly smaller otherwise, falling toward zero as the dimension grows — the finite-sample face of the curse of dimensionality on CEM strata [6]. This is the combinatorial twin of Proposition 2: conditioning on more of the record leaves strictly fewer comparable controls.

Why MIB, and a design trade-off. The inclusion rests on MIB's *separability*: each covariate carries its own tuning parameter, and appending one adds a constraint without relaxing the bounds on the others [6]; exact matching on coarsened cells inherits this, since a new period can only split strata, never merge them. A non-MIB matcher does not nest — nearest-neighbor matching on a scalar caliper, or a propensity score re-estimated as covariates accrue, re-ranks candidates at each t, so a donor pruned early can reappear later. Two payoffs follow: the per-case count $|\mathcal{M}_i(t)|$ is a *model-free* positivity diagnostic whose last crossing of a minimum $k_{\min}$ dates the landmark, complementing the score-based reading above; and MIB bounds model dependence uniformly in t once the calipers are fixed [6], a guarantee a re-estimated score cannot give as the dimension grows. The trade-off is real, however: this clean nesting is a property of *exact* CEM on the cumulative coarsened cells, not of the large-scale propensity score of Stage 2, whose t-dependent fit re-coarsens the space and breaks the inclusion. Monotone nesting therefore argues for carrying CEM into the time-varying layer; the curse of dimensionality, which the score was introduced to tame, argues against it — a tension resolved by choosing how much of the history to match *exactly* versus *by score.*

The diagnostic cascade. The mechanism behind the collapse is the ordering policy f_O of Section 3. As the latent state approaches the event, the policy orders the very test that establishes the PSI; that measurement is a near-deterministic function of H_{it} — a *descendant of the outcome*, not a baseline confounder. Conditioning on it does two harmful things at once: it saturates e_{it} at 1 for the cases (extinguishing overlap), and, were one to match on it, it would condition on a post-event consequence and induce overconditioning/collider bias. Positivity failure here is not a nuisance to be smoothed over — it is the data reporting that no comparable control exists once the diagnosis has declared itself.

Design implication: the blanking window as the trade-off knob. This is the conceptual justification for the landmark $t_i^* = E_i - b$. The window b chooses *where on the information path to stop*: increasing b moves the landmark earlier, recovering overlap (better positivity) at the cost of weaker conditioning (more residual confounding); decreasing b tightens confounding control but pushes into the diagnostic cascade, where overlap vanishes and outcome-descendant conditioning begins. The admissible range of b is whatever leaves e_{t^*} non-degenerate — a non-empty risk set with genuine common support — and there is an interior choice, to be selected empirically rather than by convention. This is the temporal analogue of static overlap remedies: trimming units with extreme scores [15] or overlap-weighting toward the comparable subpopulation [13, 14], except that here we trim in *time* rather than across units, moving the goalposts back to the last horizon at which a counterfactual still exists in the data.

Practical reading. Positivity should therefore be diagnosed *as a function of the landmark*: estimate e_{it} at a sequence of horizons and locate the point at which the case and control score distributions begin to separate irrecoverably. The estimand of Section 4 is well posed only up to that horizon; beyond it, the apparent precision of any matching or weighting estimator is illusory, because the comparison units it relies on are extrapolations rather than observations.

6 The counterfactual-selection procedure

6.1 Stage 0 — cohort construction and data-governance filters

Before any matching, restrict $\mathcal{D}$ (and verify $\mathcal{C}$) by:

1. **Supplier exclusions.** Remove all admissions from `DATA_SUPPLIER_ID` = 1990 (Advocate Aurora) entirely; remove suppliers 3707 and 3490, whose Jan-1 placeholder dates and partial lab corruption would invalidate any time-anchored matching.
2. **Eligibility / time zero.** Keep inpatient encounters with a non-null `EN_START_DATE`; define $t = 0$ at admission, mirroring an incident (new-user) design so that all covariates used for matching are measured at or after a common, well-defined origin.
3. **Risk-set construction.** For each case landmark t_i^*, form $\mathcal{R}(t_i^*)$ = donors still admitted and event-free at stay-time t_i^* (Assumption 1).

6.2 Stage 1 — baseline coarsened exact matching with missingness-as-information

We first prune the donor pool to baseline-comparable units using coarsened exact matching (CEM), the monotonic-imbalance-bounding (MIB) method [6, 7] preferred over propensity-score matching for the baseline layer.

Step 1 (Coarsen)**.** Apply κ to each baseline variable in X_i: bin continuous variables (age, height, weight, BMI) into clinically meaningful intervals, and **add an explicit *missing* category** to each coarsened variable. The missing category implements the missingness-as-information rule (Remark 2) and follows the "missingness-incorporated-in-attributes" idea.

Step 2 (Exact-match on the coarsened key)**.** Match cases to donors that share the coarsened profile on a *compact* key: sex, broad age band, facility type, urban/rural, and admission department. Do **not** include SDOH in the exact key — it is present for a negligible fraction of admissions and would empty the strata; retain it for description only. Discard donors in no occupied stratum (automatic common-support pruning).

Step 3 (Sensitivity to missingness mechanism)**.** As a robustness check, repeat Stage 1 under multiple imputation of the baseline variables, matching *within* each imputed dataset and pooling, and compare the resulting strata to the missingness-as-information solution.

6.3 Stage 2 — time-varying risk-set matching on a balancing score

Within each baseline stratum and risk set, we now match on the evolving multimodal history. The key methodological choice is to *reduce dimension through a balancing or prognostic score* rather than through an unsupervised projection.

(a) Build the per-modality history vector at (i, t).

- **Labs (numeric, irregular, MNAR).** For each analyte, summarize its within-window trajectory by interpretable features — last value, slope, min/max, count of abnormal flags, time-since-last result — *together with* the order indicator R^{lab}_{itk}. Reserve functional PCA for densely sampled analytes only. This replaces naive PCA on raw lab matrices, which is inappropriate under irregular sampling and informative missingness.
- **Code modalities (procedures, prescriptions).** Treat each as a high-dimensional sparse code set — the high-dimensional-propensity-score (hdPS) / large-scale-propensity-score (LSPS) setting. Generate candidate covariates from each code "dimension" over the window (presence, recurrence), rather than hand-picking.
- **Diagnosis stream $\overline{\gamma}_{it}$.** Handle the diagnoses likewise as a sparse code set, but treat them with care: only entries recorded at $t \le t_i^*$ enter the feature vector, so the pre-event cascade is excluded, and the analyst should verify that no retained diagnosis code is a near-deterministic marker of the impending PSI (a descendant of the outcome rather than a confounder; Section 5).
- **Vitals.** Summarize as for labs (last value, slope, abnormal counts).

(b) Reduce via a time-varying balancing score. Fit, at each landmark, an L_1-regularized (LASSO) logistic model of case-versus-risk-set membership on the full stacked history,

$$e_t(\overline{\mathcal{I}}_{it}) \;=\; \mathbb{P}\big(i \in \mathcal{C} \,|\, \overline{\mathcal{I}}_{it},\ \mathcal{R}(t)\big), \tag{6}$$

i.e. a large-scale propensity score over tens of thousands of code- and lab-derived covariates. Optionally enrich the code representation with pretrained clinical-concept *embeddings* for semantic similarity. In parallel, fit a *prognostic* score $\psi_t(\overline{\mathcal{I}}_{it})$ = predicted control-arm risk as a function of the observed history (the empirical, observed-data counterpart of the hazard (5), estimated on the event-free donors). Matching (or weighting) on e_t controls confounding; additionally balancing on ψ_t buys bias-robustness, the double-score logic.

(c) Match within the risk set. For each case i at t_i^*, select $\mathcal{M}_i(t_i^*) \subset \mathcal{R}(t_i^*)$ within the Stage-1 stratum by nearest neighbor on $e_{t_i^*}$ (logit scale) within a caliper, k:1 **with replacement** to lower bias given the large donor pool, followed by Abadie–Imbens bias correction and matching-based variance. Because cases are rare ($n_{\mathcal{C}} = 145$), we recommend reporting, as a parallel analysis, a *weighting* estimator — overlap weights (bounded, minimum asymptotic variance) or fine stratification — which avoids discarding donors and is well behaved under extreme scores and rare outcomes.

(d) Carry forward. Use the post-landmark trajectories of $\mathcal{M}_i(t_i^*)$ as the estimated counterfactual $\widehat{\tau}_i$, and aggregate across $\mathcal{C}$ to the chosen estimand (Section 4).

6.4 Stage 3 — verification by placebo falsification

Stages 1–2 *impose* balance; Stage 3 *tests* it. The logic is falsification: if the counterfactual set was well chosen, then conditional on the information used to choose it, case status carries no further signal, and a flexible learner — here a causal forest [30, 31] — should be unable to detect any effect on an outcome whose true effect is zero. "No detectable effect" is the *pass* condition; a detected effect is residual selection.

(a) Setup and the placebo outcome. Within a PSI condition type, let the pseudo-treatment be case membership, $W_i = \mathbb{1}\{i \in \mathcal{C}\}$, let X be the information conditioned on — $\overline{\mathcal{I}}_{i,0}$ to audit the baseline layer (Stage 1) or $\overline{\mathcal{I}}_{i,t_i^*}$ to audit the landmark match (Stage 2) — and let Y^{pl} be a *placebo* outcome whose effect is zero by construction. Two kinds qualify: a baseline variable *held out* of the matching (a lab, a comorbidity flag), which is pre-determined so that case status cannot cause it; or a *negative-control* outcome later in the stay with no plausible link to the PSI pathway (Section 7). The PSI label itself must **not** serve as Y^{pl}: with $Y \equiv W$ the estimated effect is identically one and tests nothing. Under the selection-on-observables condition (Assumption 3),

$$\tau^{\mathrm{pl}}(x) \;=\; \mathbb{E}\big[Y^{\mathrm{pl}}(1) - Y^{\mathrm{pl}}(0) \mid X = x\big] \;=\; 0 \qquad \text{for all } x,$$

so any departure of the fitted $\widehat{\tau}^{\mathrm{pl}}$ from zero is attributable to residual imbalance, not to a causal effect.

(b) The null, read three ways. Fit $\texttt{causal\_forest}(X, Y^{\mathrm{pl}}, W)$ on the matched set and require all three to be null: (i) the doubly-robust average effect (the AIPW estimand) has a confidence interval that brackets zero; (ii) the test for *any* heterogeneity — the best-linear projection of the conditional effect, or the differential-prediction/calibration test of Chernozhukov et al.'s generic-ML inference [32] — is non-significant, i.e. the forest finds no region of covariate space in which cases and controls diverge on the placebo; and (iii) across a panel of placebo outcomes the p-values are approximately Uniform$(0, 1)$, with a spike near zero flagging selection.

(c) A power contrast: raw pool versus matched set. Failing to reject is not evidence of good selection unless the test could have detected bad selection. We therefore run the identical procedure twice — on the *raw* donor pool and on the *matched* counterfactual set. On the raw pool the forest should recover a large, heterogeneous pseudo-effect (cases differ markedly from unselected admissions); on the matched set it should collapse toward zero. The contrast simultaneously demonstrates that the test has power and that the selection step removed the detectable imbalance.

(d) Reporting and cautions. Report *equivalence-style* evidence — the width of the interval around zero against a pre-specified negligible-effect margin — rather than bare $p > 0.05$, since a null can reflect low power. Power is the binding constraint: with $n_{\mathcal{C}} = 145$ split across condition types, a per-type forest is starved and will report "no effect" for the wrong reason. We therefore pool across types, entering condition type as a covariate so the forest borrows strength, and give per-type results descriptively with honestly wide intervals. Finally, a verification run on the same covariates and score used for matching is partly mechanical; the held-out and negative-control placebos of step (a) are what make the test informative rather than circular. Stage 3 is thus a localized, heterogeneity-aware complement to the global negative-control calibration of Section 7.

7 Diagnostics, calibration, and sensitivity

- **Balance.** Report standardized mean differences for all baseline and time-t features before and after selection; target a stringent threshold across the high-dimensional set, as in large-scale studies.
- **Negative controls + empirical calibration.** Run the full pipeline on dozens of exposure/outcome pairs with no plausible relationship to estimate the residual systematic-error distribution, then calibrate p-values and confidence intervals.
- **Sensitivity to unmeasured confounding.** Report an E-value (and/or Rosenbaum bounds) for the headline contrast.
- **Placebo-in-time and blanking-window sweeps.** Re-estimate with shifted landmarks and with $b \in \{3, 6, 9\}$ grid steps to check that results are not driven by the event prodrome.

8 Positioning relative to existing designs

The procedure sits in the EHR branch of the reviewed literature: CEM for the baseline layer (rather than propensity-score *matching*, per the King–Nielsen caution); hdPS/LSPS for the code modalities; risk-set / incidence-density matching at a landmark for the temporal alignment; and OHDSI-style negative-control calibration for residual bias. If the analytic target is the effect of a specific care *action* on PSI risk, replace "case membership" in (6) by the action indicator and adopt the active-comparator, new-user contrast; the remaining machinery is unchanged. Synthetic-control methods are not appropriate here because the units are micro (patients), not aggregate.

9 Open problems and limitations

1. **Unmeasured latent state.** Conditioning on the observed history $\overline{\mathcal{I}}_{it}$ rather than H_{it} leaves residual confounding; the diagnostics above bound but do not remove it.
2. **Positivity in high dimensions.** As the code/lab feature set grows, strict overlap is harder to satisfy; overlap weighting and principled trimming are partial answers.
3. **Time-varying confounding.** If treatment strategies are themselves of interest, sequential matching should be complemented by g-methods (marginal structural models, the g-formula) on top of comparison-observation selection.
4. **Blanking window b.** The 24h buffer trades reverse-causation protection against information loss; report it as a sensitivity parameter.

5. **Interference.** Within-unit dependence (staffing, outbreaks) can violate SUTVA; consider facility-time strata.
6. **Rare events.** With 145 cases, variance dominates; favor k:1-with-replacement matching plus bias correction, or weighting, over 1:1 matching.

10 Concluding remarks

The thread running through the paper is that selecting comparison observations for a patient-safety event is a *design* problem disciplined by how the data are generated. The structural model of the diagnostic process explains why the record is a selected, action-dependent projection of a latent state, and therefore why comparison units must be chosen on the evolving information set $\overline{\mathcal{I}}_{it}$ and reduced through a balancing rather than a variance-maximizing score. The dynamic-positivity result sharpens the usual overlap caveat into a statement about *time*: the counterfactual exists in the data only up to the horizon at which the case and donor score distributions separate, which is what the blanking window protects. The procedure operationalizes all of this — governance and quality filters, missingness-as-information baseline matching, and a landmarked risk-set match on a large-scale propensity / prognostic score — and the diagnostics state what must be reported to make the resulting counterfactuals credible. The natural next step is empirical implementation on the OMNY cohort, estimating e_{it} across a sequence of landmarks to locate the horizon at which positivity is lost and reporting the calibrated, sensitivity-bounded contrasts that follow.

References

[1] Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *J. Educ. Psychol.* 66(5), 688–701.

[2] Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika* 70(1), 41–55.

[3] Rosenbaum, P. R., & Rubin, D. B. (1984). Reducing bias in observational studies using subclassification on the propensity score. *JASA* 79, 516–524.

[4] Stuart, E. A. (2010). Matching methods for causal inference: a review and a look forward. *Statistical Science* 25(1), 1–21.

[5] Imbens, G. W., & Wooldridge, J. M. (2009). Recent developments in the econometrics of program evaluation. *J. Econ. Lit.* 47, 5–86.

[6] Iacus, S. M., King, G., & Porro, G. (2011). Multivariate matching methods that are monotonic imbalance bounding. *JASA* 106(493), 345–361.

[7] Iacus, S. M., King, G., & Porro, G. (2012). Causal inference without balance checking: coarsened exact matching. *Political Analysis* 20(1), 1–24.

[8] King, G., & Nielsen, R. (2019). Why propensity scores should not be used for matching. *Political Analysis* 27(4), 435–454.

[9] Abadie, A., & Imbens, G. W. (2006). Large sample properties of matching estimators. *Econometrica* 74(1), 235–267.

[10] Abadie, A., & Imbens, G. W. (2011). Bias-corrected matching estimators. *JBES* 29(1), 1–11.

[11] Hirano, K., Imbens, G. W., & Ridder, G. (2003). Efficient estimation of ATE using the estimated propensity score. *Econometrica* 71(4), 1161–1189.

[12] Hainmueller, J. (2012). Entropy balancing for causal effects. *Political Analysis* 20(1), 25–46.

[13] Li, F., Morgan, K. L., & Zaslavsky, A. M. (2018). Balancing covariates via propensity score weighting. *JASA* 113(521), 390–400.

[14] Li, F., Thomas, L. E., & Li, F. (2019). Addressing extreme propensity scores via the overlap weights. *AJE* 188(1), 250–257.

[15] Crump, R. K., Hotz, V. J., Imbens, G. W., & Mitnik, O. A. (2009). Dealing with limited overlap in estimation of average treatment effects. *Biometrika* 96(1), 187–199.

[16] Hansen, B. B. (2008). The prognostic analogue of the propensity score. *Biometrika* 95(2), 481–488.

[17] Li, Y. P., Propert, K. J., & Rosenbaum, P. R. (2001). Balanced risk set matching. *JASA* 96(455), 870–882.

[18] Langholz, B., & Goldstein, L. (1996). Risk-set sampling in epidemiologic cohort studies. *Statistical Science* 11(1), 35–53.

[19] Hernán, M. A., & Robins, J. M. (2016). Using big data to emulate a target trial. *AJE* 183(8), 758–764.

[20] Hernán, M. A., Sauer, B. C., Hernández-Díaz, S., Platt, R., & Shrier, I. (2016). Specifying a target trial prevents immortal time bias. *J. Clin. Epidemiol.* 79, 70–75.

[21] Ray, W. A. (2003). Evaluating medication effects outside of clinical trials: new-user designs. *AJE* 158(9), 915–920.

[22] Lund, J. L., Richardson, D. B., & Stürmer, T. (2015). The active comparator, new user study design. *Curr. Epidemiol. Rep.* 2, 221–228.

[23] Schneeweiss, S., et al. (2009). High-dimensional propensity score adjustment. *Epidemiology* 20(4), 512–522.

[24] Tian, Y., Schuemie, M. J., & Suchard, M. A. (2018). Evaluating large-scale propensity score performance. *IJE* 47(6), 2005–2014.

[25] Schuemie, M. J., Ryan, P. B., DuMouchel, W., Suchard, M. A., & Madigan, D. (2014). Interpreting observational studies: why empirical calibration is needed. *Stat. Med.* 33(2), 209–218.

[26] Schuemie, M. J., et al. (2020). LEGEND: assessing validity using hypertension. *JAMIA* 27(8), 1268–1277.

[27] Leyrat, C., et al. (2019). Propensity score analysis with partially observed covariates: multiple imputation. *IJE* (and related work, Mitra & Reiter, 2016).

[28] VanderWeele, T. J., & Ding, P. (2017). Sensitivity analysis in observational research: the E-value. *Ann. Intern. Med.* 167(4), 268–274.

[29] Chernozhukov, V., et al. (2018). Double/debiased machine learning. *Econometrics Journal* 21(1), C1–C68.

[30] Wager, S., & Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *JASA* 113(523), 1228–1242.

[31] Athey, S., Tibshirani, J., & Wager, S. (2019). Generalized random forests. *Ann. Statist.* 47(2), 1148–1178.

[32] Chernozhukov, V., Demirer, M., Duflo, E., & Fernández-Val, I. (2018). Generic machine learning inference on heterogeneous treatment effects in randomized experiments. *NBER Working Paper 24678.*